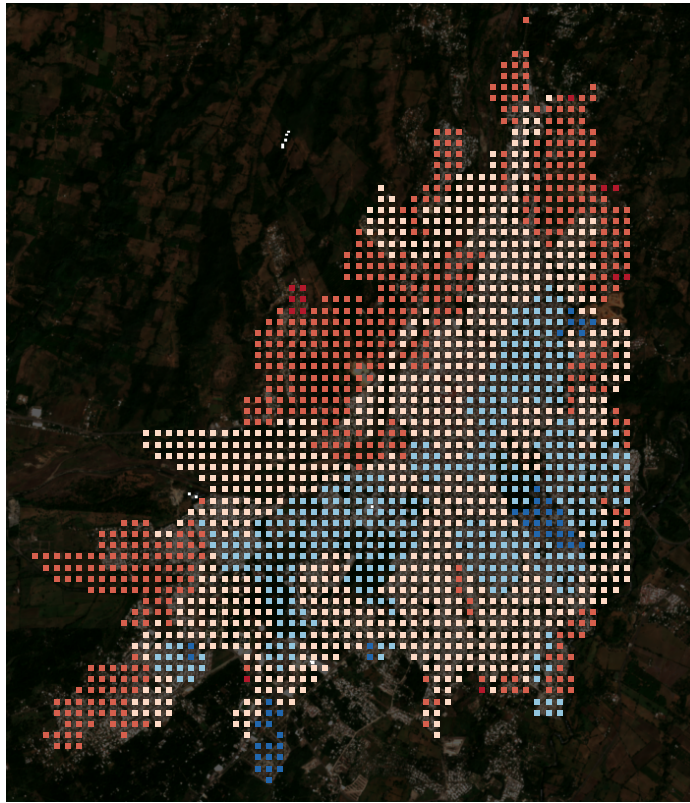


Supplementary Information

Municipal aggregates recover ninety per cent of the neighbourhood deprivation ordering that their satellite features support

AGEB ground truth (held out)



Supplementary Figure 1: Tract-level ground truth for Tapachula, on the colour scale of main-text Fig. 1c. Every token inherits the held-out GRS grade of the AGEB it falls in.

Supplementary Table 1: The supervision-efficiency curve (main-text Fig. 2a). Mean over three seeds on the 14 validation cities; s.d. over seeds.

Bags	Within- ρ	s.d.	Map AUROC	Bag MAE
50	0.150	0.021	0.789	0.138
100	0.141	0.021	0.792	0.130
200	0.151	0.022	0.784	0.115
400	0.182	0.020	0.778	0.102
771	0.227	0.010	0.802	0.098

Supplementary Table 2: Agreement by tract size (main-text Limits). Within-municipality Spearman over municipalities with at least ten tracts in the bin.

Tokens per tract	Tracts	Median population	Municipalities	Within- ρ
1–3	567	211	15	0.257
4–9	849	1,948	16	0.338
10–19	999	2,644	20	0.352
20–49	464	3,231	14	0.432
≥ 50	79	3,561	2	0.127

Supplementary Table 3: Municipalities excluded from the expansion for insufficient optical depth or persistent acquisition errors; the guard requires eight observations per pixel and excludes rather than composites shallow ground.

Municipality	Reason
Amozoc; Comalcalco; Hidalgo del Parral; Lagos de Moreno; Nuevo Casas Grandes	composite below the eight-observation depth floor (optical arm)
La Independencia	inconsistent tract geometry
El Marqués	repeated acquisition timeouts